

SHAKYA BHATTACHARYYA

[Email](#) | (585) 305-5583 | [LinkedIn](#) | [Website](#) | New York, NY (ready to relocate)

CPT/OPT work authorization valid through Dec 2028

AI & Data Scientist | Machine Learning · Predictive Modeling · NLP

Data Scientist specializing in predictive modeling, credit risk, and pricing analytics. Skilled in Python, SQL, and TensorFlow; delivered \$30M+ cost avoidance, 25% model lift, and 15% retention improvement by deploying models that directly shaped business decisions.

Core Competencies: Machine Learning · NLP · Predictive Modeling · Experiment Design (A/B) · Forecasting · SQL · Python

PROFESSIONAL EXPERIENCE

Simon Vision Consulting | Rochester, NY

Oct 2024 - Dec 2024

Data Science Analyst (Client Engagement: City of Rochester)

- Built supervised learning retention models (Python, scikit-learn); improved AUC +0.07 and enabled a 15% retention lift.
- Designed and executed A/B tests with SQL/Python; validated interventions using uplift modeling and cross-validation.
- Applied NLP (TF-IDF, topic modeling) on survey data to identify dropout drivers, informed city-wide funding and strategic design.

Cognizant Technology Solutions | Kolkata, India

Sep 2022 - Jun 2024

Business Data Analyst (ML Focus)

- Engineered forecasting and cost-risk ML models (regression, XGBoost, Python, SQL) on 250K+ trial records; reduced decision turnaround 20% and achieved \$30M+ cost avoidance.
- Automated ETL pipelines and batch ML workflows (SQL, Airflow); cut reporting cycles 90% and delivered predictive outputs for global leadership.
- Deployed TensorFlow models for trial risk scoring; improved detection precision +12% and decision speed.
- Optimized feature engineering and ML pipelines, boosting efficiency 20% and reducing latency across production runs.
- Partnered with stakeholders to translate predictive modeling into strategy for resource allocation and fraud-risk mitigation.

IEM-IEDC (Govt. of India funded) | Kolkata, India

Feb 2019 - Dec 2021

Data Science Intern

- Improved model accuracy by 25% on 100K+ records using TensorFlow and scikit-learn for applied AI research.
- Applied NLP methods on case studies and delivered 10+ analytics workshops to 1,000+ learners on SQL, Python, and Tableau.

EDUCATION

Simon Business School, University of Rochester | Rochester, NY

Jul 2024 - Dec 2025

Master of Science in Business Analytics (STEM) | GPA: 3.9/4.0

- Advanced Certificate in Pricing Analytics (Revenue Optimization Focus)
- Relevant Coursework: Machine Learning, NLP, Deep Learning, Predictive & Prescriptive Analytics, Forecasting

Maulana Abul Kalam Azad University of Technology | Kolkata, India

Aug 2018 - Jun 2022

Bachelor of Technology in Computer Science & Engineering | GPA: 8.24/10

- Director's Award for Overall Student Performance

SKILLS & CERTIFICATIONS

Technical: Python (pandas, NumPy, scikit-learn, TensorFlow) · SQL · Tableau · Power BI · Excel

AI/ML: Predictive Modeling · NLP · Forecasting · A/B Testing · Credit Risk Modeling · ETL

Certifications: Google Data Analytics Professional Certificate (2025)